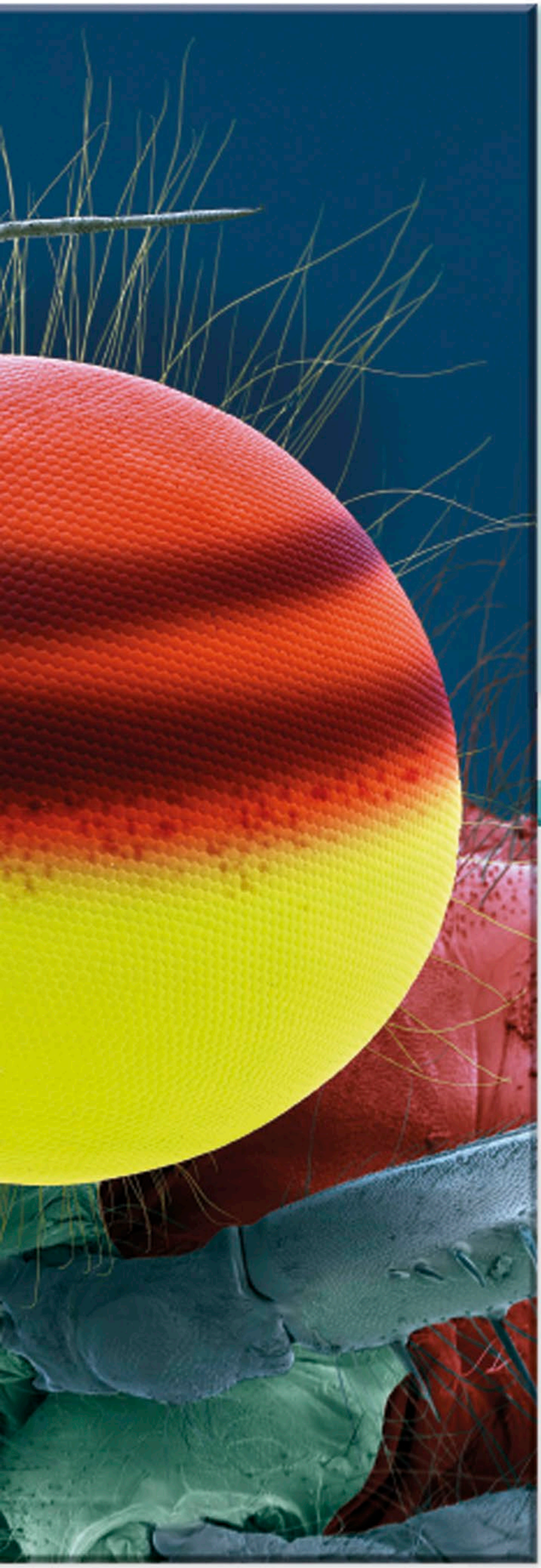

Zooming in on the details

The compound eyes of a large red damselfly loom large in this compelling close-up image. Such images were made possible by the development of the scanning electron microscope (SEM) from the 1930s onward. SEMs scan a specimen, often held in a vacuum, using an extremely narrow beam of electrons to trace over the object. These microscopes can achieve much higher magnifications (50–100,000 times) and resolutions than optical microscopes, which use light magnified by lenses. Objects measured in nanometers (billionths of a meter) can be imaged clearly by SEMs, making them invaluable for forensics and investigating tiny creatures, new drugs, and materials in incredible detail.

“Our work, it seems to me, can bring us a special bonus of pleasure and satisfaction, through… our ability to peep into the inexhaustible world of the smallest forms of existence.”

Ernst Ruska, inventor of the first electron microscope, 1958